

Dang Thi Thanh Huyen

Date of birth: March 11th, 2003

Gender: Female

Phone Number: +84 337054082

Email address: huyen.dang1103@gmail.com

Level: Undergraduate student

Major: Biomedical Engineering

University: Hanoi University of Science and Technology

CPA: 3.58/4



ABOUT MYSELF

I am an undergraduate student, majoring in Biomedical Engineering, Advanced Program (Full English Program). In the program, I took two AI-related courses which are Programming Language and Machine Learning in Healthcare. With the desire to understand more deeply about Machine Learning, I have pursued knowledge on my own through the Internet. Since my second year, I have joined a lab on AI in medical image processing. I have participated in projects to improve my technical skills and research the application of AI in image analysis. I am looking for a job to gain experience in practical projects, I can connect with AI experts and experience a professional working environment.

EDUCATION

- Bien Hoa High School for the Gifted (Major in Mathematics) 2018 – 2021 (Ha Nam)
- Hanoi University of Science and Technology 2021 – 2025

Relevant courses:

- Basic and core courses: Programming Language, Machine Learning in Healthcare (achieve A or higher)
- Mathematics: Algebra, Calculus I, Calculus II (achieve B or higher)

Certificates:

- “STUDENT WITH FIVE GOOD CRITERIA” University level, 2022 – 2023
- University President's Certificate for Contributions to the University, 2022 – 2023

PUBLICATION

- Accepted paper “Height Measurement of Recumbent Individuals in 2D Space” at 24th ICCRI (June, 2024).
- Accepted oral presentation “Controllable Medical Image Generation Using Text Prompts on Multi-Phase Contrast CT Data” at IEEE R10 HTC (Humanitarian Technology Conference) 2025

SKILLS

Technical

- Languages: Python, Latex, C++
- Tools: Github, Visual Studio Code, Jupyter notebook, Weight&Bias
- Framework: Pytorch, Scikit-learn, Tensorflow
- Data/Matrix: Numpy, Pandas, Matplotlib
- Computer Vision: CNNs, U-Net Variants, Transformer

Interpersonal

- Languages: Vietnamese (native); English (second language, VSTEP B2)
- Soft Skills: Teamwork, Technical Writing (in English), Reading Documents/Papers (in English), Presentation (in English), Active Listening, Self-study

PROJECTS

Height Measurement of Recumbent Individuals in 2D Space (2023 – 2024)

- Teamwork, subject requirement
- A method to measure the height of patients in the lying position in the hospital, using images of the lying patient.
- Tasks: Data collection, data processing, evaluating, paper and report writing
- Tools: Mediapipe, OpenCV

Keypoint Detection (2024)

- Personal, practice
- Detect 5 keypoints's coordinates on a face (using CelebA dataset)
- Tasks: Organize code, data preprocessing
- Tools: Pytorch, Albumentations, Visual Studio Code
- Achievement: Loss approximately 0.0001 for the dataset with around 130,000 images

Overlapped Chromosomes Segmentation (2023 – 2024)

- Teamwork, subject requirement
- A method to detect overlapped chromosomes (using images taken in metaphase)
- Tasks: Data processing, training and evaluating, poster and report writing
- Tools: Pytorch, Wandb, Visual Studio Code
- Achievement: separate overlapping zones with an overlap detection accuracy of 79.21% and foreground accuracy of 96.08%

DecodetAll (2023)

- Teamwork, practice
- A sandbox repo for practicing implementing encoder-decoder from scratch across various domains and applications.
- Tasks: Retinal Vessel Segmentation using Unet, AttUnet, RattUnet, NestedUnet, AttUnet
- Tools: Pytorch, Jupyter notebook

Controllable text-to-image generation for CT scans at contrast enhanced phases (2024 – 2025)

- Teamwork, subject requirement, graduation thesis
- Fine-tuning the Imagen model, training model on dataset including text and image pairs with text is prompt guidance, created perspective by using metadata.
- Tasks: Data processing, create prompt, training, poster and report writing
- Tools: Pytorch, Wandb, Visual Studio Code
- Achievement: Enhance the controllable and high-resolution property in CT image generation